Inverse Design of Refractory High Entropy Alloys using CVAE

(Conditional Variational Autoencoder)

G-15 : 23254, 23248, 23244

Motivation & Problem Statement

- Designing Refractory High-Entropy Alloys (RHEAs) is challenging due to vast composition space
- Traditional trial-and-error experiments are expensive & time-consuming
- Demand for alloys with high strength at elevated temperatures in aerospace, energy, defense

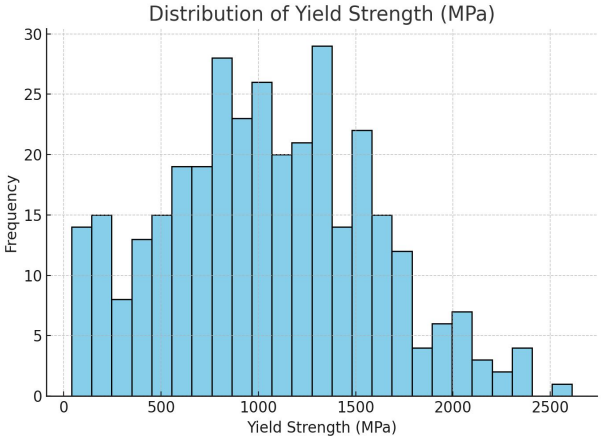
- Given a target yield strength, can we generate candidate alloy compositions?
- Need a model that:
  - Learns structure–property relationships
  - Generates valid & diverse alloy compositions
  - Provides interpretable formulas (e.g., $Al_{0.25}MoNbTiV$)

Objectives

- **Modeling goal:** Develop a Conditional Variational Autoencoder (CVAE) for the inverse design of RHEAs.
- **Specific objectives:**
  - Clean raw alloy data, extract element fractions, and encode categorical/numerical features.
  - Implement CVAE with encoder, decoder, conditional prior, and property head for yield strength alignment.
  - Optimize reconstruction + Kullback-Leibler (KL) divergence + property prediction losses.
  - Sample candidate alloys for a target yield strength and decode compositions.
  - Compare generated alloys against target values and analyze their elemental composition.

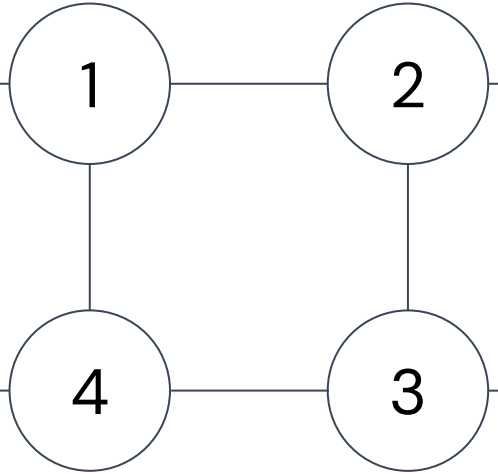
Dataset Overview

- **Source:** Couzinié et al., Data in Brief, 2018 (Comprehensive compilation of mechanical properties for refractory HEAs/RCCAs)
- **Size:**
  - 122 alloys; 255 unique (comp. + T) pairs
  - 340 mechanical test records (2010–2018)
- **Attributes Recorded:**
  - Alloy compositions (mole fractions of elements)
  - Metallurgical state (as-cast, annealed, HIP, etc.)
  - Phase content (single vs multiphase)
  - Testing mode (tension / compression)
  - Temperature range: −268 °C to 1600 °C
  - Properties: Density (ROM), Young's modulus (exp./ROM), Yield strength, Specific strength



- **Value:**
  - Provides a verified benchmark for comparing model-generated predictions
  - Enables mapping of composition → properties relationships

Pipeline



Data Preparation

- Alloy compositions → normalized fractions.
- Cleaned numeric props, encoded categoricals, scaled features.

CVAE Architecture

- Encoder: features + target → latent space
- Conditional prior & decoder: reconstruct alloy features.
- Property head: predict yield strength

Candidate Generation

- Input target yield → refine latent vector → generate alloys.
- Evaluate by closeness to target & element composition.

Training

Reconstruction + Kullback-Leibler (KL) divergence + property loss.

Data Preprocessing

Cleaning Num. Values

- Handled missing values, inconsistent formats, RT → 25 °C
- Extracted valid numbers from ranges/notations

Composition Encoding

- Parsed chemical formulas into elemental fractions (Al_frac, Mo_frac, etc.)
- Normalized to ensure fractions sum to 1

Categorical Encoding

- Phase conditions & test settings → LabelEncoder (e.g., Equilibrium, Single/Multiphase)

Feature Scaling

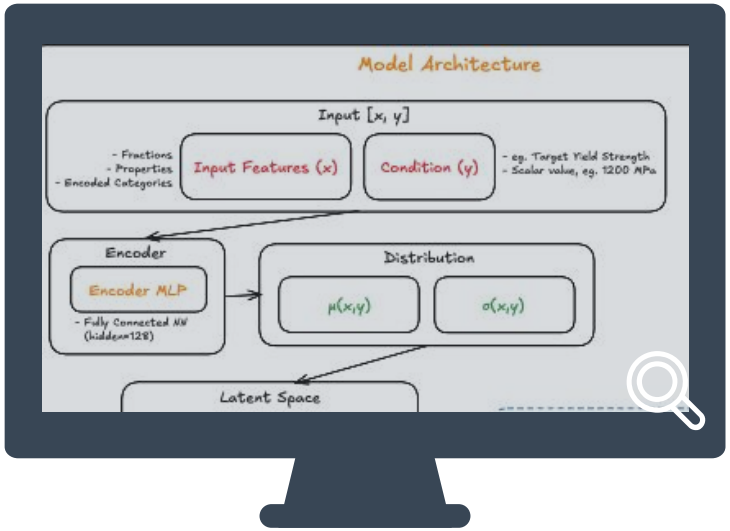
- StandardScaler for features (X)
- MinMaxScaler for target (Yield Strength)

Train–Validation Split

- 80/20 split, stratified on target distribution

Model Architecture

Click on the computer screen to access the link for model architecture.



Training Setup

Goal: learn conditional generative model $p_\theta(x \mid y)$ with latent z aligned to property y.

$$\mathcal{L} = \underbrace{\mathrm{MSE}(\hat{x}, x)}_{\text{Reconstruction}} + \beta \underbrace{D_{KL}(q_\phi(z|x,y) \parallel p_\psi(z|y))}_{\text{KL Divergence}} + \lambda_{\text{prop}} \underbrace{\mathrm{MSE}(\hat{y}, y)}_{\text{Property Prediction}}$$

where, **β warm-up:** increases linearly from 0 → 2.0 over first 100 epochs; **λ_{prop}:** fixed at 0.5

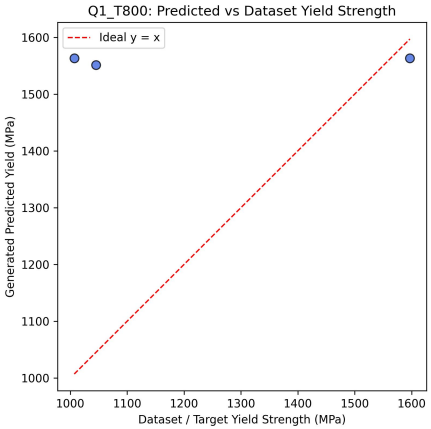
Optimiser	AdamW	**Epochs (max)**	200	**Hidden units**	256 (e/d), 64 (pr, ph)
Learning Rate	1e-3	**Early Stopping**	Patience = 40	**Activation**	Gaussian Error Linear Unit (GELU)
Weight Decay	1e-5	**Gradient Clipping**	1.0	**Split**	80/20 (train/val)
Batch Size	32 / 64	**Latent dim. z**	8	**Scaling**	Std. (x), MinMax (y)

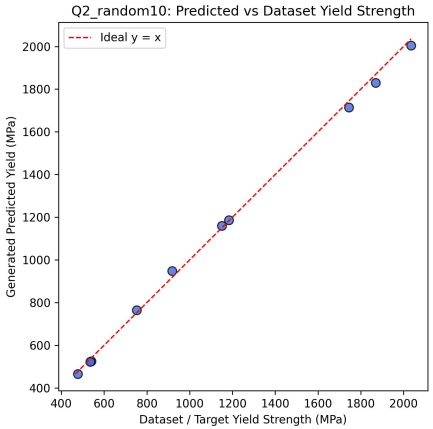
Model Interpretation & Visualisation Techniques

t-SNE (t-Distributed Stochastic Neighbor Embedding)	PFI (Permutation Feature Importance)	Correlation Heatmap
• Reduces high-dimensional latent vectors (z) into 2D. • Nearby points = alloys with similar compositions or properties. • Reveals clusters for high/low strength and temperature.	• Randomly shuffles each feature and measures performance drop. • Larger drop = higher feature importance. • Identifies which elements most affect yield strength.	• Compares feature–strength correlations between the training dataset and generated alloys. • High similarity = model preserves real metallurgical trends.

Quantitative Model Verification

Validate that the CVAE predicts realistic yield strengths for both known and unseen alloy conditions.





Test Case	Regime	MAE (MPa)	± 5 % Match	R^2	Remarks
Q1 (800 °C)	Extrapolation	**365**	33 %	0.87	Valid high-T predictions
Q2 (Dataset)	Interpolation	**19**	100 %	0.99	Accurate reconstruction

Key Findings

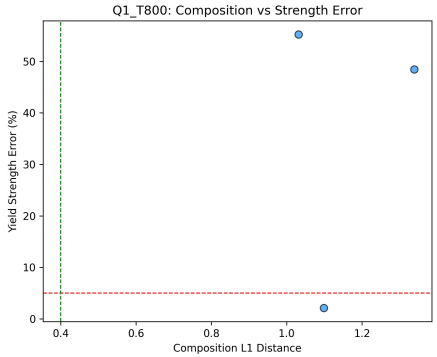
- Interpolation (Q2): < 3 % error → excellent property fidelity.
- Extrapolation (Q1 @ 800 °C): Predicts plausible high-T strengths (~1.5 GPa) and valid refractory chemistries despite ~20 % numerical deviation.
- Overall: $R^2 \approx 0.99$ → strong trend-level agreement between composition and yield strength.

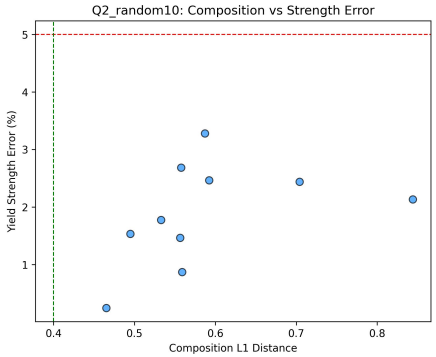
Interpretation

Points near y = x show close dataset agreement; high-T points remain within the realistic strength band.

Qualitative Model Verification

Examine whether generated alloys are metallurgically valid despite compositional differences.





Scatter plots show strength error vs composition difference — most points remain below ±5 % strength deviation despite large compositional shifts.

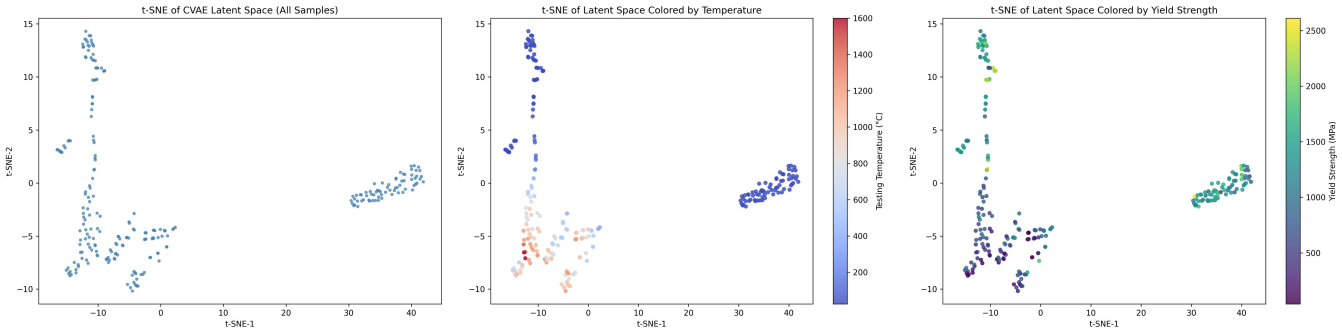
Key Findings

- Generated alloys differ ($L_1 \approx 0.5–1.0$) yet keep yield strengths within ±5 %.
- CVAE learns multiple valid chemistries for similar properties.
- Substitutions (Hf→Cr/W, Mo→Nb/Ta) match known refractory strengthening trends.
- Indicates generalization, not memorization.

Case	Dataset Formula	Generated Formula	ΔStrength	Comment
Q1 @ 800 °C	$Hf_{0.17}Mo_{0.17}Nb_{0.17}Ta_{0.17}Ti_{0.17}Zr_{0.17}$	$Cr_{0.16}Nb_{0.25}Ta_{0.18}Ti_{0.15}V_{0.14}W_{0.13}$	+ ≈ 550 MPa	Hf/Mo → Cr/W; same strength range
Q2 (25 °C)	$Al_{0.10}Nb_{0.20}Ta_{0.16}Ti_{0.30}V_{0.04}Zr_{0.20}$	$Al_{0.11}Cr_{0.11}Nb_{0.24}Ti_{0.23}V_{0.16}Zr_{0.15}$	– ≈ 30 MPa	Ti ↔ Cr shift; strength retained

Latent Space Analysis

Visualize how the CVAE organizes alloys by yield strength and temperature in latent space.



t-SNE visualization shows that the model's latent space clusters alloys by strength and temperature, proving learned physical structure.

Key Observations

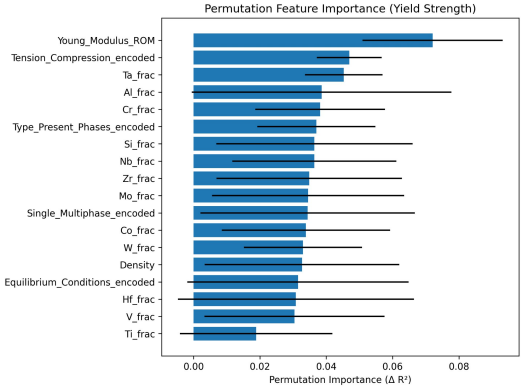
- Distinct clusters emerge → alloys grouped by composition similarity.
- Smooth color gradients for yield strength → confirms continuous property encoding.
- Temperature-based separation → strong conditioning; latent space preserves thermomechanical behavior.
- No random scatter → model has learned a structured representation, not noise.

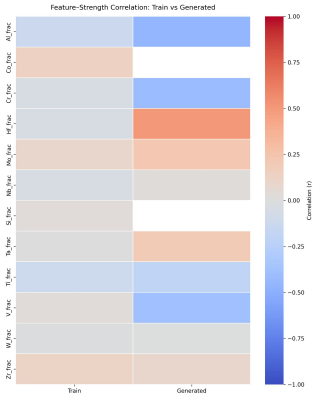
Interpretation

- Latent dimensions capture composition–property relationships.
- High-strength and high-T alloys form identifiable regions.
- Confirms that latent space is physically meaningful and interpolative.

Explainability & Correlation

Assess which features drive model predictions and whether generated alloys follow dataset trends.





Interpretation

- PFI confirms CVAE focuses on physically meaningful alloy features.
- Correlation map shows generated alloys statistically mirror training data.
- Results validate interpretability + trustworthy strength prediction.

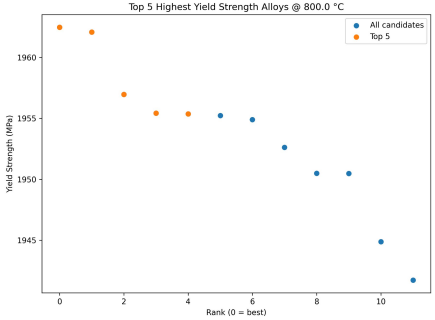
Key Findings

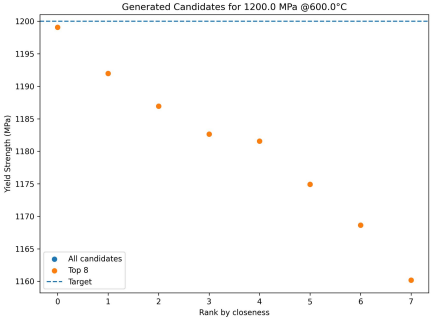
- Top influential: Ta, Al, Cr, Young_Modulus_ROM, Tension_Compression_encoded → key drivers of yield strength.
- Stable trends: Mo, Hf, Ta retain similar correlations (train vs generated) → confirms data fidelity.
- Low-impact: Co, V, Ti show weak or variable effects.
- Overall: Model learns realistic composition–strength relations consistent with metallurgy.

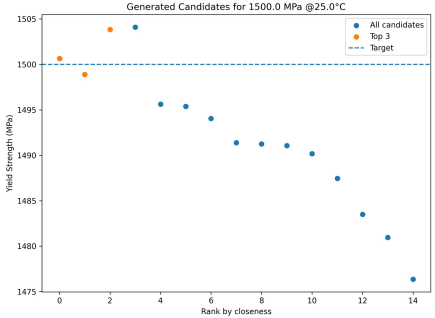
Model explanations align with metallurgical knowledge — confirming both interpretability and realism in generated alloys.

Generative Alloy Design Using CVAE

The CVAE enables targeted alloy design — generating new compositions that meet desired yield-strength objectives







- Generated alloys are novel yet physically consistent with dataset trends.
- Confirms that the CVAE can be used as a data-driven alloy recommender for targeted performance.

Future Work

Multi-Property Conditioning

- Extend CVAE to predict multiple mechanical properties (e.g., hardness, elastic modulus) simultaneously.
- Enables trade-off optimization between strength, ductility, and stability.

Local Explainability

- Apply Integrated Gradients (IG) to explain individual alloy predictions.
- Identify which elements contribute most to a specific alloy's yield strength.

Experimental Validation

- Validate generated alloys through CALPHAD, DFT, or experimental synthesis.
- Confirms physical feasibility beyond statistical similarity.

Advanced Generative Models

- Compare CVAE with diffusion models or conditional GANs for property-controlled generation.
- Explore hybrid models combining CVAE + Physics-informed Neural Networks (PINNs).

Conclusion

- The Conditional VAE successfully models composition–property relationships for refractory HEAs.
- Achieved high property accuracy ($R^2 \approx 0.99$) and physically valid generalization to unseen alloys.
- Latent space and explainability confirm learned physical structure, not memorization.
- Generated alloys show realistic, metallurgically consistent compositions with targeted yield strengths.
- The framework demonstrates a data-driven pathway for inverse design in materials discovery.

Thanks!